



Retrieval practice with feedback facilitates the acquisition of English spelling rules and etymological knowledge in EFL learners

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Abstract

This study investigated whether retrieval practice with feedback enhances elementary EFL learners' acquisition of English spelling knowledge—a critical yet challenging component of L2 writing proficiency. Although spelling instruction in many EFL contexts relies heavily on rote memorization, retrieval with feedback may improve learning of morphophonological spelling rules and etymological knowledge, two areas where effective instructional approaches have been underexamined. Here, etymological knowledge refers to instructionally presented word-origin and word-structure associations, including simplified information about roots, affixes, lexical stems, place-name elements, and semantic or cultural associations. In two within-subjects experiments, one of which was preregistered, fourth-grade EFL learners acquired English spelling patterns via retrieval practice (application questions with corrective feedback) or traditional reading-based study, with retention tested at 24 h and one week. Experiment 1 ($n = 86$) targeted 20 spelling rules, whereas Experiment 2 ($n = 84$) featured 16 instructionally presented etymological associations. Retrieval practice with feedback improved learning in both experiments, including superior performance at 24 h ($d = 0.47$) and one week ($d = 0.70$) for spelling rules, with performance remaining stable across both time points, and consistently higher retention than reading at both 24 h ($d = 0.33$) and 7 days ($d = 0.50$) for etymological knowledge, with performance again remaining stable across both time points. Thus, retrieval practice with feedback appeared to benefit learning of spelling rules and etymological knowledge targeted in the present intervention, as reflected in multiple-choice application performance after the tested delays. Implications for L2 literacy instruction and practical recommendations are discussed.

Keywords English spelling rules · Etymology · Retrieval practice · Reading practice · Testing effects · EFL teaching methods



Introduction

English spelling poses formidable challenges for learners worldwide, yet it remains an overlooked component of second language (L2) vocabulary instruction (Ehri & Rosenthal, 2007). This neglect is particularly consequential for learners whose first language (L1) employs a fundamentally different writing system, such as speakers of non-alphabetic languages acquiring English as a foreign language (EFL) (Bear et al., 2018). Although considerable attention has been devoted to teaching word meanings and pronunciation, the systematic instruction of spelling patterns—including morphophonological rules and etymological knowledge—has received limited empirical investigation in L2 contexts. Etymological knowledge is defined here as instructionally presented word-origin and word-structure associations, including simplified information about roots, affixes, lexical stems, place-name elements, and semantic or cultural associations. The present study addresses this gap by examining whether retrieval practice with feedback, a powerful learning strategy established in cognitive psychology, can enhance elementary EFL learners' acquisition of English spelling rules and etymological knowledge.

Spelling as a critical component of L2 vocabulary: challenges for EFL learners

Within Nation's (2012) influential framework of word knowledge, spelling is a vital part of productive vocabulary, enabling learners to use words accurately in writing. However, unlike receptive recognition, productive spelling requires precise knowledge of letter sequences, morphological patterns, and the systematic relationships between sounds and orthography (Schmitt, 2014). For EFL learners, spelling accuracy directly impacts writing quality, academic assessment, and communicative competence (Figueredo, 2006; Graham et al., 2002). Despite its importance, spelling instruction in many EFL contexts remains anchored in rote memorization approaches that fail to develop learners' understanding of underlying patterns, thereby limiting the efficiency of their spelling acquisition (Al-Yahmedi et al., 2019).

Learners whose L1 is morphosyllabic face particular challenges in acquiring English spelling competence due to profound differences between their L1 and English writing systems. For example, Chinese employs a morphosyllabic system where characters represent morphemes rather than phonemes, whereas English uses an alphabetic system encoding phonological information (Wang et al., 2003). This fundamental difference means that Chinese learners must develop entirely new forms of metalinguistic awareness to process English orthography effectively (Guan et al., 2011). Specifically, such learners must acquire: (1) phoneme-grapheme correspondence rules absent in their L1, (2) morphological spelling patterns (e.g., doubling consonants before *-ing*), and (3) etymological knowledge that explains seemingly irregular spellings (see also Ye et al., 2023). Without explicit instruction, learners from non-alphabetic L1 backgrounds often rely on visual memorization strategies adopted from their first writing system—an approach that proves inefficient for English's morphophonological system (Zhang & Koda, 2012).

Current practices and retrieval practice with feedback in EFL spelling instruction

Spelling instruction in many EFL contexts, particularly in Asia, emphasizes repetitive copying and whole-word memorization, reflecting broader tendencies toward rote learning (Hu, 2003). Although in alignment with traditional literacy practices, this approach fails to leverage the systematic nature of English orthography. It also contrasts with arguments that effective spelling instruction should be based on aspects of spelling knowledge such as rules and etymology rather than memorization alone. Indeed, some researchers argue that visual memorization strategies are inadequate for English spelling, which requires not just memorization of individual spellings but also an understanding of sound-letter correspondences and morphological patterns encompassing numerous words (Joshi et al., 2008).

Insufficient training in spelling rules is one of the major factors contributing to spelling errors among EFL learners (Albeshier, 2018). Even when instruction in spelling rules is provided, it often relies on passive exposure through reading or teacher explanations, with limited opportunities for active application (for related discussion see Pan et al., 2021). This pedagogical approach contradicts robust findings from cognitive science showing that active retrieval promotes deeper learning than passive study (Carpenter et al., 2022; McDermott, 2021; Roediger & Butler, 2011). The mismatch between instructional practices and learning science principles may partially explain why spelling remains a persistent challenge for EFL learners throughout their academic careers.

In addition, etymological knowledge remains an underused resource in L2 spelling instruction. English orthography is not simply a set of arbitrary spellings; it reflects phonological, morphological, and historical layers of the language. For learners, accurate and developmentally appropriate information about word origins can make some otherwise opaque spelling patterns more meaningful. For example, in many Greek-derived words, the sound /f/ is spelled *ph*, as in *photosynthesis* and *philodendron*, and /k/ may be spelled *ch*, as in *chlorophyll* and *chemistry* (Joshi et al., 2008). More broadly, spelling instruction that attends to sound, meaning, morphology, and word origin can support learners' analytic understanding of English spelling rather than encouraging reliance on visual memorization alone (Joshi et al., 2008; Venezky, 1999). In EFL contexts, recent work has also suggested that etymology-based instruction may support vocabulary learning, although available evidence remains limited and has focused mainly on adult learners rather than children's spelling development (Osman, 2023). Effective methods for teaching etymological knowledge to young EFL learners remain underexplored.

Retrieval practice—the learning technique which involves actively recalling information from memory—has emerged as one of the most effective learning strategies across diverse domains (Dunlosky et al., 2013). The *retrieval practice effect*, demonstrated in hundreds of studies (Pan et al., 2024), shows that testing enhances long-term retention more than restudying, even when initial retrieval attempts are effortful or unsuccessful (Rowland, 2014). Importantly, retrieval practice benefits extend beyond rote memorization to support meaningful learning and transfer (Carpenter, 2012; Pan & Rickard, 2018). Correct answer feedback after a retrieval attempt is

known to substantially enhance the effects of retrieval practice (Rowland, 2014). Recent meta-analyses indicate that retrieval practice yields substantial benefits in authentic educational contexts (Adesope et al., 2017), yet its application to L2 spelling instruction remains largely unexplored. The few studies examining retrieval practice in L2 contexts have focused primarily on vocabulary meaning rather than orthographic knowledge (Karatas et al., 2025; Nakata, 2017), and within the latter, spelling rules or etymological knowledge have not been the focus. This gap is particularly striking given that spelling rules and etymological patterns represent exactly the kind of systematic knowledge that presumably could benefit from active retrieval and application.

Theoretical framework: extending retrieval practice with feedback to metalinguistic knowledge

The present study extends research on retrieval practice by testing its effectiveness for two types of spelling knowledge that have received minimal attention: (1) morphophonological spelling rules and (2) etymological knowledge. This extension is theoretically significant because both knowledge types are abstract and rule-based rather than item-specific, requiring learners to extract patterns and apply them to novel words.

We propose that retrieval practice with feedback should be particularly effective for learning spelling patterns because it promotes several cognitive processes crucial for L2 orthographic development: (1) Active generation: Attempting to apply spelling rules during retrieval requires deeper processing than passive recognition (Morris et al., 1977); (2) Error correction: Retrieval attempts followed by feedback can make gaps in learners' understanding of rules more salient, thereby facilitating targeted remediation (Metcalfe, 2017); (3) Transfer-appropriate processing: Practice applying rules to novel words mirrors authentic spelling demands (Morris et al., 1977); and (4) Desirable difficulties: The effort required during retrieval strengthens memory traces—that is, learning conditions which are initially more challenging can yield better results over the long term (Bjork & Bjork, 2011). Furthermore, for learners whose L1 involves holistic character recognition, retrieval practice may facilitate the analytic processing necessary for understanding English morphophonological patterns. We also hypothesize that retrieval practice will be effective for etymology learning among EFL students because it requires learners to actively connect word origins to spelling patterns—a form of elaborative processing that should enhance retention.

The present study

In two experiments, we investigated whether retrieval practice with feedback supports the acquisition of English spelling rules and etymological knowledge in elementary EFL learners, two domains not yet studied within the retrieval practice literature. Experiment 1 examined the learning of morphophonological spelling

rules (e.g., dropping silent 'e' before adding *-ing*), whereas Experiment 2 focused on etymological knowledge (e.g., Greek and Latin roots). Appendices A and B catalogue the different types of rules and knowledge learned across experiments. Both experiments compared retrieval practice with corrective feedback against traditional reading-based study, measuring retention after 24 h and one week. By testing fourth-grade EFL students who have received basic English instruction but lack explicit spelling rules or etymological knowledge, we examined whether young EFL learners can benefit from metalinguistic instruction delivered through retrieval practice. This population is theoretically interesting because they are developing L2 literacy skills while still solidifying L1 literacy—a critical period for cross-linguistic awareness development (Koda, 2007).

Our research addressed three key questions:

1. Can retrieval practice with feedback enhance elementary EFL learners' acquisition of English spelling rules compared to traditional reading-based study?
2. Does retrieval practice with feedback similarly benefit learning of instructionally presented word-origin and word-structure associations?
3. How stable are these benefits over a one-week retention interval?

To address these questions, we conducted two classroom-based experiments with fourth-grade EFL learners in China. Both experiments employed within-subjects designs where participants learned different sets of target knowledge through both retrieval practice with feedback and reading, allowing direct comparison of instructional methods while controlling for individual differences. In Experiment 1, students learned 20 English spelling rules divided into two sets, with each set taught through a different method. Experiment 2 followed the same design but focused on 16 etymological concepts about English word origins. Retrieval practice consisted of application-based multiple-choice questions with corrective feedback, whereas the reading condition involved studying explanatory texts for an equivalent duration. Learning outcomes were assessed through application tests requiring students to generalize their knowledge to novel English words, providing a stringent measure of meaningful learning rather than rote memorization. The experiments were conducted with different classes, providing an initial opportunity to examine whether similar patterns of results would emerge across different types of spelling knowledge and different groups of students within the same educational context.

Experiment 1

The first experiment investigated whether retrieval practice with feedback—actively recalling and applying information from memory, then receiving the correct answers as feedback—would enhance elementary EFL learners' acquisition of English spelling rules compared to traditional reading-based study, among native Chinese speakers as measured on retention tests conducted after delays of one day and one week.

Method

Participants

An a priori power analysis using G*Power 3.1.9.7 (Faul et al., 2007) was conducted to determine the sample size needed to detect an effect size of Cohen's $d = 0.50$ (for reference, Jones et al. (2016) reported an advantage of $d = 0.60$ for retrieval practice over copying among elementary school students learning English spelling words). The analysis indicated that 40 participants would provide over 80% power to detect such an effect on a two-tailed t -test with $\alpha = .05$. The final sample consisted of 86 fourth-grade students (42 boys, 44 girls; mean age = 9.63 years) from two classes in a primary school in Shandong Province, China. This sample size well exceeded our preregistered target of 40 participants (approximately the size of each class), providing sufficient power for the primary analyses and allowing for data exclusions (ultimately, there was no need to exclude any participants).

All participants were native Mandarin speakers. Starting in the third grade, these students had taken regular English second language classes each semester, which by that point had equipped them with knowledge of the English alphabet and elementary-level vocabulary. Crucially, they had not received any prior instruction in English spelling rules and had minimal-to-no explicit knowledge of those rules, making them highly suitable for the present research. Teachers asserted that their cognitive and behavioral abilities were adequate for the tasks.

The entire study (Experiments 1 and 2), which was approved by the Science and Technology (Medicine) Ethics Committee at Tsinghua University (THU01-20240166), took place in November 2024 with voluntary participation and guardian consent.

Design

The experiment used a within-subjects design where each student experienced both learning methods (retrieval practice with feedback and reading) with different spelling patterns. We tested retention at two intervals (24 h and 7 days) to examine both immediate and delayed learning outcomes—timeframes relevant to typical EFL instruction cycles. The two classes learned English spelling rules through retrieval practice with feedback and reading, respectively, in an order that was counterbalanced across classes. All of the spelling rules had not previously been taught in the classes. Participants took a retention test assessing spelling rule knowledge on the second and seventh days after instruction had concluded. The dependent variable was the number of questions answered correctly on the tests.

Materials

The materials consisted of two sets of English spelling rules. For each set, there was one in-class practice test (for retrieval practice) and one set of reading materials (for reading practice). Additionally, two retention tests were developed for each set of spelling rules to assess learning outcomes.

Spelling rules. The materials consisted of 20 common English spelling rules that EFL learners must master for accurate writing. These rules included morphological patterns (rules for how word parts combine, such as adding *-ing* or *-ed*), silent letter patterns, and vowel-consonant combinations that often confuse Chinese learners due to L1-L2 differences, all drawn from English language textbooks by Carney (1994) and Stormonth (2022). The rules, which are detailed in Appendix A, were chosen with the guidance of two primary school English teachers, who confirmed that the difficulty level was appropriate for fourth-grade students. Although the participants had not yet received formal instruction in these rules, they would be introduced in the following school years as part of the English language curriculum for Grades 5 and 6. The rules were divided into two sets—Set A and Set B—each comprising 10 rules.

Retrieval practice materials. Two 10-item practice tests were developed for the retrieval practice condition, each corresponding to one set of spelling rules (Set A or Set B). All items were multiple-choice application questions with four response options, with each item addressing a single rule. An example item (translated fully into English) was, “How do you spell “hope” in the progressive tense? A. hopping, B. hopeing, C. hoppng, D. hoping.” We used application questions that required students to actively recall spelling rules and apply them to new words. This approach—rather than simple recognition—mirrors authentic writing tasks where EFL learners must generate correct spellings independently. Research shows such application-based retrieval practice produces stronger learning than passive recognition (Pan & Rickard, 2018).

Reading materials. Two sets of explanatory reading materials were prepared for the reading condition (one for Set A and another for Set B). The materials always began with a list of the 5 vowel letters in English and mentioned that the remaining 21 letters were consonants. Each spelling rule was accompanied by a brief explanation and illustrative examples, formatted for clarity and printed on A4 paper. For instance, the “double consonant rule” was explained thusly (translated into English): “When a verb ends with a short vowel followed by a single consonant, the final consonant is usually doubled before adding “-ing.” For example, the progressive form of “run” is “running,” and the progressive form of “stop” is “stopping.”” Overall, the reading materials resembled the type of explanations that can be found in English language textbooks. In the reading condition, each student received an individual copy of the relevant reading materials.

Retention tests. Two retention tests were developed for each set of spelling rules. These tests did not directly assess English spelling rule knowledge specifically but rather involved the application of these rules to novel English words—an approach mirroring real-world language situations where learners must generalize and apply spelling patterns to unfamiliar vocabulary. Each test contained 10 multiple-choice questions with four response options each, with one question for each spelling rule. The questions, which were presented in a randomized order, resembled those used during practice in the retrieval practice condition but were worded differently and involved different target words than on the practice tests. All tests were designed to be similar in content, format, number of items, and difficulty.

It is acknowledged that because all ten rules were assessed on each test—and each set of rules was assessed on two separate days—results for the second of those tests

must be interpreted in light of the fact that the students had already attempted to recall and apply those same rules on an earlier test. We were aware of this limitation during study development, but the test design was nevertheless retained in order to balance practical constraints in classroom implementation with the need to obtain both short- and longer-delay assessments of performance within the same instructional cycle.

Procedure

Both classes learned the two sets of spelling rules using the same general procedures. In a within-subjects design, each class learned one set through retrieval practice with feedback and the other set through reading practice. Assignment of rule set to learning condition and the order of learning conditions were counterbalanced across the two classes. Specifically, Class 1 learned Set A through retrieval practice with feedback first and then learned Set B through reading practice, whereas Class 2 learned Set A through reading practice first and then learned Set B through retrieval practice with feedback. Thus, each rule set appeared in both learning conditions across classes, and each learning condition appeared in both ordinal positions. Once both the instructional session and learning session (retrieval practice or reading) for the first set had been completed, students immediately began the instructional session for the next set. To measure learning outcomes, retention tests were administered 24 h and one week later. To control for time-on-task, the retrieval practice and reading activities were time-matched within each class; students spent 15.23 min and 14.92 min on the learning activities in the first and second classes, respectively.

Instructional sessions. An experienced elementary school English teacher led instructional sessions for each set of rules, explaining the patterns and providing examples. The rules and examples were presented on slides closely aligned with the reading materials. Teachers prepared an outline in advance to ensure consistency in content and session length within each series of rules. Each instructional session lasted approximately 25 min, with a research assistant timing the sessions and providing reminders when time was up.

Retrieval practice with feedback. In the retrieval practice condition, after the instructional session, students completed a 10-item, pen-and-paper practice test covering the rules that had just been taught. After attempting all of the questions, the students reviewed and corrected their mistakes under the teacher's guidance. Specifically, for each test question, the teacher provided correct answer feedback (via projected slides and verbal narration) that included the correct answer and an explanation of the relevant spelling rule. A research assistant monitored the time taken and gave the instructor reminders to ensure that the appropriate amount of time was taken.

Reading. In the reading condition, after the instructional session, students were given the relevant reading materials for the appropriate rule set and required to read it carefully. A research assistant monitored the time taken and gave the instructor reminders to ensure that the appropriate amount of time was taken.

Retention tests. The retention tests, which occurred in the same classroom and were untimed, were conducted on the following day and one week later.

Scoring and data analysis

Given that all retention test items involved multiple-choice format, there was no ambiguity in the scoring. Each test item was worth one point, and the total score for a single retention test was 10 points. Although the preregistered analysis plan specified using conventional Analysis of Variance (ANOVA) to analyze total scores, we instead used a generalized linear mixed-effects model with a logistic link function (which uses a logit link to model log-odds of accuracy) as it enables modeling accuracy at the item level. Item-level accuracy was analyzed using a generalized linear mixed-effects model with a binomial error distribution and logit link, fit in R using the *glmer()* function from the *lme4* package. The model included fixed effects of Learning Method, Retention-Test Delay (Day), and their interaction, as well as covariates for academic record (standardized) and class, and random intercepts for participants and items: ($correct \sim LearningMethod * Day + class_id + academic_z + (1 | student_id) + (1 | item)$). Learning Method was coded with retrieval practice as the reference level and reading as the comparison level; Retention-Test Delay was coded with 24 h as the reference level and 7 days as the comparison level. Marginal contrasts were estimated using the *emmeans* package. Cohen's *d* values were calculated from participant-level proportion-correct difference scores as the mean paired difference divided by the standard deviation of the paired differences. The model converged without convergence or singularity warnings. The primary and sensitivity GLMMs (the latter excluding class as a covariate) were fit using default *glmer()* optimizer settings, whereas supplementary class-moderation models used the *bobyqa* optimizer ($maxfun = 200,000$). The updated analysis scripts, rerun HTML outputs, and session information from a reproducibility rerun in R 4.5.1, including software and package versions, are available on OSF. For transparency, the originally preregistered ANOVAs on total scores were also conducted. These ANOVAs, which yielded comparable patterns of results, are presented in the Supplementary Materials.

Transparency and openness

This experiment's basic design, sampling procedures, and analysis plan were preregistered at <https://aspredicted.org/n6dj-gvkc.pdf>. Materials, data, and analysis code are archived at: <https://osf.io/6bnkx/>. Experiment 1 deviated from the preregistered protocol in several respects; for a table featuring quotes from the preregistration and associated changes, in line with Willroth and Atherton (2024), please see the Supplementary Materials. The final sample included 86 participants rather than the preregistered target of 40; retention tests contained 10 multiple-choice items per test instead of 8 items; and retention intervals were implemented as 24 h and 7 days rather than 24 h and 5 days. Further, analyses were conducted at the item level using binary accuracy rather than aggregated test scores. As noted above, the preregistered repeated-measures ANOVA was replaced as the primary analysis by a generalized linear mixed-effects model implemented in R, which included covariates and random effects for participants and items. Overall, these deviations reflected methodological refinements made to better match the structure of the collected data and did not alter the substantive design logic of the experiment.

Results

Descriptive performance during the initial retrieval practice phase is reported first to characterize students' early application of newly learned spelling rules. Subsequent analyses examine the effects of retrieval practice versus traditional reading-based study on elementary EFL learners' retention and application of English spelling rules, assessed after a 24-h delay and after a one-week delay.

Retrieval practice learning performance

During the initial retrieval practice with feedback phase, students demonstrated their ability to apply newly learned spelling rules to novel English words—a challenging task for EFL learners who typically rely on memorization. The first class correctly answered 0.53 of items ($SE = 0.03$), while the second class achieved 0.67 accuracy ($SE = 0.03$). This level of accuracy exceeded 0.50, a threshold associated with stronger retrieval practice effects (Rowland, 2014). These results indicate that elementary students could begin applying abstract spelling patterns even during initial learning. This moderate success level is pedagogically important, suggesting the task was appropriately challenging without being overly frustrating, falling within students' zone of proximal development.

Retention test performance

Descriptive statistics for retention test performance are presented in Table 1. We fit a generalized linear mixed-effects model to examine the effects of Learning Method (retrieval practice with feedback versus reading), Retention-Test Delay (24 h versus 7 days), and their interaction on item-level accuracy. Crossed random intercepts for participants and items were included to account for individual differences and item variability. Academic record (students' school academic-performance score; included as a standardized covariate to control for baseline differences in general academic aptitude) and class were included as covariates. In the following results, descriptive means and standard errors are calculated from participant-level proportion correct scores, averaged across items; inferential statistics for Learning Method, Retention-Test Delay, and their interaction are based on marginal contrasts from the fitted model, whereas effects of covariates (academic record and class) are taken directly from the model coefficients. Further, β represents the effect of a term on the

Table 1 Class-wise descriptive statistics for retention test performance from Experiment 1

Class	Learning method	Delay	<i>M</i>	<i>SE</i>
First	Retrieval Practice with Feedback	24 h	0.58	0.03
		7 days	0.56	0.03
	Reading	24 h	0.54	0.03
		7 days	0.41	0.03
Second	Retrieval Practice with Feedback	24 h	0.60	0.04
		7 days	0.58	0.04
	Reading	24 h	0.44	0.03
		7 days	0.34	0.03

log-odds of a correct response; positive values indicate higher odds, and negative values indicate lower odds.

In the generalized linear mixed-effects model, there were significant main effects of Learning Method and Retention-Test Delay, as well as a significant Learning Method by Retention-Test Delay interaction. Based on marginal contrasts derived from that model (see Table 2, row 1), overall item-level accuracy was higher following retrieval practice with feedback ($M = 0.58$, $SE = 0.02$) than reading ($M = 0.43$, $SE = 0.02$), $\beta = 0.69$, $p < 0.001$. There was also a significant main effect of Retention-Test Delay (see Table 2, row 2), with item-level accuracy higher at the 24-h test ($M = 0.54$, $SE = 0.02$) than at the 7-day test ($M = 0.47$, $SE = 0.02$), $\beta = 0.32$, $p < 0.001$. Academic record was positively associated with item-level accuracy, $\beta = 0.37$, $p < 0.001$, whereas class did not significantly predict performance ($\beta = -0.21$, $p = 0.14$). A model including the Learning Method $\times$ Retention-Test Delay $\times$ Class interaction did not provide clear evidence of class-based moderation, $X^2(1) = 0.04$, $p = 0.838$. Given that only two classes were included, this result should however be interpreted cautiously. Descriptively, the class-wise means and standard errors showed a broadly similar pattern across classes (see Table 1), but these descriptive patterns should not be interpreted as decisive evidence of generalizability beyond these classes.

With respect to the Learning Method by Retention-Test Delay interaction, there was a significant interaction, $\chi^2(1) = 8.58$, $p = 0.003$, indicating that the advantage of retrieval practice with feedback over reading increased over time (see Table 2, rows 3–5 for the simple-effect contrasts). That is, the two methods appear to have led to different forgetting patterns—a crucial consideration for EFL contexts where review time is limited.

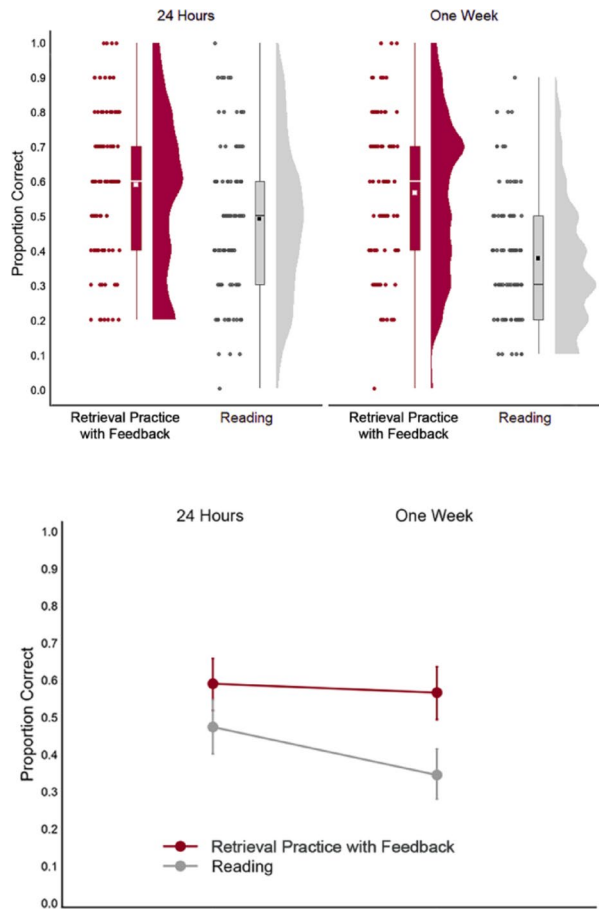
As depicted in Fig. 1, the participant-level distributions (upper panel) and the model-based estimates (lower panel) show the same general pattern: reading-based learning showed the typical forgetting curve familiar to language teachers—with performance declining from 24 h ($M = 0.49$, $SE = 0.02$) to one week ($M = 0.38$, $SE = 0.02$), whereas retrieval practice with feedback remained relatively stable, with students maintaining similar performance at 24 h ($M = 0.59$, $SE = 0.02$) versus one week ($M = 0.57$, $SE = 0.02$). These results indicate higher delayed multiple-choice

Table 2 Model-based pairwise contrasts for retention test performance from Experiment 1

Comparison	β (log odds)	SE (β)	95% CI (β)	p	d
Retrieval Practice with Feedback versus Reading	0.69	0.07	[0.54, 0.84]	< 0.001	0.68
24 h. versus 7-days	0.32	0.07	[0.17, 0.46]	< 0.001	0.49
Retrieval Practice with Feedback 24-h versus Reading 24-h	0.47	0.10	[0.20, 0.74]	< 0.001	0.47
Retrieval Practice with Feedback 7-days versus Reading 7-days	0.91	0.11	[0.64, 1.18]	< 0.001	0.70
Retrieval Practice with Feedback 24-h versus Retrieval Practice with Feedback 7-days	0.10	0.10	[-0.17, 0.37]	0.781	0.11

Results derived from the generalized linear mixed-effects model reported in the text. β values represent log-odds. Cohen's d reflects within-subject effect sizes from participant-level difference scores. Rows 1–2 report planned contrasts (unadjusted); rows 3–5 report Tukey-adjusted simple effects

Fig. 1 Spelling rule test performance from Experiment 1. *Note.* The upper panel presents raincloud plots of participant-level mean proportion-correct scores for each retention test. Each point represents an individual student's overall test performance at the corresponding delay, illustrating the distribution of observed scores across participants. The lower panel presents model-based predicted probabilities of a correct response derived from the generalized linear mixed-effects model, with error bars representing standard errors (overlap between error bars does not necessarily indicate non-significant differences between conditions). The model estimates reflect item-level accuracy while accounting for repeated measures across participants and items



application-test performance following retrieval practice with feedback than reading across the tested intervals.

A supplementary model excluding class as a covariate produced the same pattern of results based on marginal contrasts derived from the model: overall accuracy remained higher for retrieval practice with feedback than reading ($\beta = 0.69$, $p < 0.001$) and higher at 24 h than at 7 days ($\beta = 0.32$, $p < 0.001$). Importantly, the Learning Method by Retention-Test Delay interaction remained significant ($\chi^2(1) = 8.55$, $p = 0.003$). Academic record continued to positively predict item-level accuracy ($\beta = 0.37$, $p < 0.001$) based on the model coefficients. These results were consistent with the earlier findings.

Discussion

In Experiment 1, retrieval practice with feedback produced higher performance than reading on multiple-choice application tests of English spelling rules after 24 h and one week. Performance in the retrieval-practice-with-feedback condition did not significantly decline across the tested interval, whereas reading-based performance

declined more clearly. These findings suggest a potential retention advantage on the measured application-test outcomes. For EFL contexts with limited review time, this performance pattern may be pedagogically relevant, although its implications require further study.

Experiment 2

The second experiment investigated the relative benefits of retrieval practice with feedback versus reading for acquiring instructionally presented word-origin and word-structure associations. These materials are referred to below as etymological knowledge in a pedagogical sense, but they should not be understood as formal or comprehensive etymological analysis.

Method

Participants

Experiment 2 was conducted at the same school and followed the same ethics approval and informed consent procedures as Experiment 1. Two additional fourth-grade classes participated, comprising a total of 84 students (42 students in each class; 46 boys, 38 girls; mean age = 9.55 years)—a number comparable to the 86 participants in Experiment 1 and exceeding the minimum sample size determined by the power analysis conducted previously. The participants represented the entire enrollment of each class (no participants were excluded from data analysis). As in Experiment 1, the students had been systematically learning English as a second language since third grade and possessed foundational knowledge of the English alphabet, basic vocabulary, grammar, and sentence structure.

Design

Experiment 2 featured a 2 (learning method: retrieval practice with feedback versus reading) $\times$ 2 (retention-test delay: 24 h versus 7 days) within-subjects design, just as in Experiment 1.

Materials

The materials include two sets of English etymological concepts. For each set, there was one in-class practice test (for retrieval practice with feedback) and one set of reading materials (for reading practice). Additionally, two retention tests were developed for each set to assess learning outcomes.

Etymological knowledge. The target information consisted of English etymological concepts drawn from Joshi et al. (2008), presented in a pedagogically simplified instructional form designed for primary-level learners. The materials focused on selected historical roots, prefixes, and suffixes and their relevance to English spelling patterns. Simplification involved selecting a limited set of high-frequency

or instructionally salient etymological relations and presenting them as instructional generalizations rather than comprehensive historical-linguistic accounts, with developmentally appropriate explanations that did not require full etymological reconstruction. For example, learners were taught that in many Greek-derived words, the sound /f/ is represented by *ph*, as in *photosynthesis* and *philodendron*, and that /k/ may be represented by *ch*, as in *chlorophyll* and *chemistry*. Additional items highlighted common roots and affixes linked to meaning and spelling regularities. Two elementary school English teachers reviewed all instructional materials, which are detailed in Appendix B, and confirmed that the content was age-appropriate, comprehensible, and pedagogically suitable for the participants. Although some aspects of word origin are introduced in the school's upper-grade English curriculum, such instruction is typically fragmented and limited at the primary level.

Similar to Experiment 1, these materials were divided into two sets: Set A and Set B. Each set contained 8 instructional etymological concepts, for a total of 16 concepts across both sets. The items varied in the degree to which they reflected direct etymological relations, pedagogically simplified generalizations, or broader semantic or culturally associated groupings used to support instructional coherence. These materials were first presented visually through slides, accompanied by a teacher's verbal explanations. To support comprehension, visual aids—namely illustrations depicting mythological references—were included on the slides (see Fig. 2 for an example). Each instructional session introduced one set of etymological knowledge.

Retrieval practice materials. Two 8-item practice tests were developed for the retrieval practice with feedback condition, each corresponding to one etymological concept (Set A or Set B). All items were multiple-choice application questions with four response options, with each item addressing a single concept. An example item (translated fully into English) was, “Given the root ‘bio-,’ what is the most likely meaning of the word *biotech*? A. information technology B. biotechnology C. craftsman D. physical science.” As in Experiment 1, these application questions required learners to mentally retrieve the etymological concepts that they had learned and relate them to the new information presented in the question.

Reading materials. Two sets of explanatory reading materials were prepared for the reading condition (one for Set A and another for Set B). Each set consisted of 1-page expository text (approximately 1100 Chinese characters and English words) explaining eight English etymological concepts. As in Experiment 1, the reading materials resembled the types of explanations that can be found in English language textbooks. For example, one concept was described as follows (translated fully into English): “Some word roots are related to Greek mythology. Everyone knows what ‘music’ means, and words like ‘museum’ and ‘amuse’ also contain the letters ‘mus’. So, what’s the connection between them? In Greek mythology, the Muses were the goddesses of the arts and inspiration. As a result, ‘mus’ later became a root associated with entertainment. Words like ‘music’, ‘museum’, and ‘amuse’ all share the idea of using art to entertain people.” In the reading condition, each student received an individual copy of the relevant reading materials.

Retention tests. As in the first experiment, two retention tests were created for each set of etymological concepts, each requiring application of these concepts to novel words. Each test included 8 multiple-choice questions—one per concept—with four

很多词根来源于希腊语

phon 意思是“声音、噪音”，来自希腊语

- **telephone** (电话)
- **microphone** (麦克风)
- **symphony** (交响乐)

这些含有词根**phon**的单词都与声音有关

tele 表示“远处的、远程的、在远处”，也来自希腊语

- **telephone** (电话) 就是取得远处的人的声音
- **television** (电视) 就是远程看到风景



Many word roots come from Greek.

The root **phon** means "sound" or "voice"

- **tele**phon**e**
- **micro**phon**e**
- **sym**phon**y**

Words containing this root are all related to sound

The root **tele** means "far", "distant", or "remote"

- **tele**phon**e**: Hearing the voice of someone from a distance
- **tele**visi**on**: Seeing images or scenery from afar



Fig. 2 Example instructional slide for word-origin knowledge from Experiment 2. *Note.* This slide shows how some word roots were derived from Greek. The upper panel was shown to the students, who were native Mandarin speakers; the lower panel is the slide translated entirely into English for clarification purposes

response options. Questions were randomized and resembled those from the retrieval practice tests but used different wording and target words. All tests were matched in content, format, number of items, and difficulty (with all concepts tested on each of the two tests for a given set, and hence, subject to the same interpretive considerations owing to repeated testing as in Experiment 1).

Procedure

The procedure closely followed that of Experiment 1, with the primary difference being that the learning and testing materials focused on word-origin and word-structure knowledge rather than morphophonological spelling rules. On the first day, participants completed either an in-class practice test (retrieval practice with feedback) or reading practice, both after an initial instructional session lasting approximately 25 min. The practice test period also incorporated instructor-provided correct answer feedback for each question, accompanied by explanations, just as in Experiment 1. Retention tests were administered 24 h and 7 days later to assess learning outcomes. A counterbalanced design, similar to Experiment 1, was used. Specifically, one class learned Set A through retrieval practice with feedback and Set B through reading practice. The other class learned Set A through reading practice and Set B through retrieval practice with feedback. Thus, each concept set appeared in both learning conditions across classes, and each learning condition appeared in both ordinal positions. The time students spent during retrieval practice and reading practice was equated (14.46 min and 14.71 min for the first and second classes, respectively).

Scoring and data analysis

Scoring procedures were identical to Experiment 1. Retention-test accuracy was analyzed using mixed-effects models: item-level accuracy (0/1) was analyzed with a logistic mixed-effects model including fixed effects of Learning Method, Retention-Test Delay, and their interaction, with random intercepts for participants and items, and academic record and class as covariates. Item-level accuracy was analyzed using a generalized linear mixed-effects model with a binomial error distribution and logit link, fit in R using the *glmer()* function from the *lme4* package. The model formula was: $correct \sim LearningMethod * Day + class_id + academic_z + (1 | student_id) + (1 | item)$. Learning Method was coded with retrieval practice as the reference level and reading as the comparison level; Day was coded with Day 1 (24 h) as the reference level and Day 7 (one week) as the comparison level. Academic record was standardized before analysis. The model included 2,688 observations from 84 students and 16 items. Marginal contrasts were estimated using *emmeans*. Cohen's *d* values were calculated from participant-level proportion-correct difference scores as the mean paired difference divided by the standard deviation of the paired differences. The model converged without reported convergence or singularity warnings. The primary and sensitivity GLMMs (the latter excluding class as a covariate) were fit using default *glmer()* optimizer settings, whereas supplementary class-moderation models used the *bobyqa* optimizer ($maxfun = 200,000$). The updated analysis scripts, rerun HTML outputs, and session information from the reproducibility rerun in R 4.5.1, including software and package versions, are available on OSF. Inferential statistics for Learning Method, Retention-Test Delay, and their interaction are reported as marginal contrasts from the fitted model, whereas effects of covariates (academic record and class) are taken directly from the model coefficients.

Transparency and openness

This experiment was not specifically preregistered, but followed the same overall design, sampling approach, and analysis plan as that of Experiment 1. Data, materials, and code are archived as described previously.

Results

Similar to the preceding experiment, descriptive performance during the initial learning phase is reported first to characterize students' early acquisition of the word-origin and word-structure associations taught in Experiment 2. Subsequent analyses examine the effects of retrieval practice with feedback versus traditional reading-based study on retention, assessed after a 24-h delay and after a one-week delay.

Retrieval practice learning performance

Students showed strong initial success in applying instructionally presented word-origin and word-structure associations during retrieval practice with feedback. The first class correctly answered 0.74 of items ($SE = 0.03$), whereas the second class also achieved 0.74 accuracy ($SE = 0.03$), indicating that EFL learners can connect word origins to spelling patterns when given the opportunity for active application. This high success rate—nearly three times chance level—suggests that etymological knowledge may be more accessible to EFL learners than assumed, especially when instruction moves beyond passive memorization.

Retention test performance

Descriptive statistics for retention test performance are presented in Table 3. Analysis followed the same procedure as described for Experiment 1, using a generalized linear mixed-effects model with item-level accuracy as the outcome, including crossed random intercepts for participants and items, and academic record and class as covariates. Descriptive statistics are participant-level means and standard errors, and inferential statistics are based on marginal contrasts from the fitted model.

Based on marginal contrasts derived from the model (Table 4, rows 1–2), overall item-level accuracy was higher following retrieval practice with feedback than reading ($\beta = 0.51, p < 0.001$), whereas there was no overall effect of Retention-Test Delay

Table 3 Class-wise descriptive statistics for retention test performance from Experiment 2

Class	Learning method	Delay	<i>M</i>	<i>SE</i>
First	Retrieval Practice with Feedback	24 h	0.69	0.04
		7 days	0.67	0.04
	Reading	24 h	0.54	0.04
		7 days	0.49	0.03
Second	Retrieval Practice with Feedback	24 h	0.65	0.03
		7 days	0.63	0.03
	Reading	24 h	0.58	0.03
		7 days	0.57	0.03

Table 4 Model-based Pairwise Contrasts for Retention Test Performance from Experiment 2

Comparison	β (log odds)	SE (β)	95% CI (β)	p	d
Retrieval Practice with Feedback versus Reading	0.51	0.09	[0.34, 0.68]	< .001	0.52
24 h versus 7-days	0.03	0.09	[-0.14, 0.20]	.741	0.04
Retrieval Practice with Feedback 24-h versus Reading 24-h	0.37	0.12	[0.06, 0.69]	.011	0.33
Retrieval Practice with Feedback 7-days versus Reading 7-days	0.65	0.12	[0.33, 0.96]	<.001	0.50
Retrieval Practice with Feedback 24-h versus Retrieval Practice with Feedback 7-days	-0.11	0.12	[-0.43, 0.21]	.818	0.10

Results derived from the generalized linear mixed-effects model reported in the text. β values represent log-odds. Cohen's d reflects within-subject effect sizes from participant-level difference scores. Rows 1–2 report planned contrasts (unadjusted); rows 3–5 report Tukey-adjusted simple effects

($\beta = 0.03$, $p = 0.741$). The Learning Method $\times$ Retention-Test Delay interaction was not significant, $\chi^2(1) = 2.51$, $p = 0.113$, suggesting that the retrieval advantage did not reliably differ across delays. Tukey-adjusted simple-effect contrasts confirmed that retrieval outperformed reading at both 24 h ($\beta = 0.37$, $p = 0.011$) and 7 days ($\beta = 0.65$, $p < 0.001$; Table 4, rows 3–4), and that retrieval performance did not significantly change from 24 h to 7 days ($\beta = -0.11$, $p = 0.818$; Table 4, row 5). Consistent with these effects, participant-level means showed higher performance for retrieval than reading at both 24 h (retrieval: $M = 0.66$, $SE = 0.02$; reading: $M = 0.56$, $SE = 0.02$) and one week (retrieval: $M = 0.65$, $SE = 0.03$; reading: $M = 0.53$, $SE = 0.02$). These patterns are depicted in Fig. 3, with the upper panel showing participant-level score distributions and the lower panel showing model-based predicted probabilities.

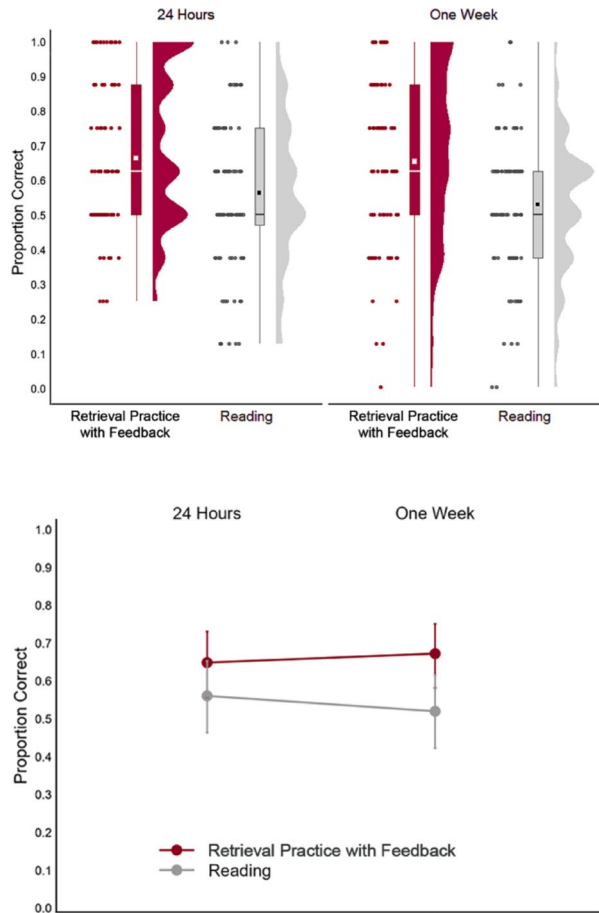
Academic record was a significant positive predictor of accuracy ($p < 0.001$), whereas class did not significantly predict performance ($p = 0.64$). In addition, a model including the Learning Method $\times$ Retention-Test Delay $\times$ Class interaction did not provide clear evidence of class-based moderation ($\chi^2(1) = 0.25$, $p = 0.620$). Given that only two classes were included, however, this result should be interpreted cautiously rather than as strong evidence of robustness beyond those classes. Descriptively, class-wise means and SE s showed broadly similar patterns across both classes (see Table 3).

A supplementary model excluding class as a covariate produced the same pattern of results based on marginal contrasts derived from the model: overall accuracy remained higher for retrieval practice with feedback than reading ($\beta = 0.51$, $p < 0.001$), whereas the effect of retention-test delay remained non-significant ($\beta = 0.03$, $p = 0.741$). The Learning Method $\times$ Retention-Test Delay interaction remained non-significant ($\chi^2(1) = 2.51$, $p = 0.113$). Academic record continued to positively predict item-level accuracy ($\beta = 0.33$, $p < 0.001$) based on the model coefficient.

Discussion

The results of Experiment 2 extend the benefits of retrieval practice with feedback to instructionally presented word-origin and word-structure associations. Retrieval practice with feedback produced higher overall learning than reading, with the advantage evident at both 24 h and one week and with no reliable evidence that

Fig. 3 Etymological knowledge test performance from Experiment 2. *Note.* The upper panel presents raincloud plots of participant-level mean proportion-correct scores for each retention test. Each point represents an individual student's overall test performance at the corresponding delay, illustrating the distribution of observed scores across participants. The lower panel presents model-based predicted probabilities of a correct response derived from the generalized linear mixed-effects model, with error bars representing standard errors (overlap between error bars does not necessarily indicate non-significant differences between conditions). The model estimates reflect item-level accuracy while accounting for repeated measures across participants and items



the retrieval advantage differed across delays. Practically, these findings suggest that retrieval practice with feedback may be useful for supporting learners' delayed application of instructionally presented word-origin and word-structure associations on classroom-style assessment tasks. Whether these benefits extend to productive spelling or authentic writing tasks remains an important question for future research.

General discussion

The present research provides evidence that retrieval practice with feedback can support elementary EFL learners' delayed application of English spelling rules and word-origin knowledge relative to reading-based study. Across two experiments, we demonstrated that elementary students who engaged in retrieval practice with feedback consistently outperformed those who studied through traditional reading, as evident on multiple-choice application tests administered after 24 h and one week. These findings contribute to research on L2 orthographic learning and suggest a potential classroom approach for helping learners apply spelling-related knowledge after instruction.

Theoretical contributions: extending the boundaries of retrieval practice

Our results extend retrieval practice research by demonstrating benefits for delayed multiple-choice application-test performance involving metalinguistic knowledge. Previous research has predominantly examined retrieval practice with discrete factual information or vocabulary meanings (Rowland, 2014). By showing consistent benefits for both morphophonological spelling rules and instructionally presented word-origin and word-structure associations, we provide evidence that retrieval practice with feedback can support delayed performance on multiple-choice application tests involving novel items. These findings are consistent with the possibility that several features of the retrieval-practice-with-feedback condition—including effortful retrieval, corrective feedback, active application, and practice–assessment alignment—jointly supported delayed application-test performance.

The advantages of retrieval practice with feedback we observed (Cohen's d ranging from 0.33 to 0.70 for the retrieval-versus-reading contrasts at each delay) are comparable to those reported in meta-analyses of retrieval practice research (Rowland, 2014), despite our use of retention tests requiring application to novel words rather than verbatim recall. This finding suggests that retrieval practice with feedback may support delayed performance when learners must apply their knowledge to novel items, aligning with Carpenter's (2012) framework on transfer of learning following retrieval practice. Moreover, it is also in alignment with Pan and Rickard's (2018) framework which highlights the role of application-based questions as a form of “elaborated retrieval practice” that promotes more flexible learning. From a transfer-appropriate processing perspective, the retrieval-based application tasks used in the present study may have further contributed to the observed advantages because they more closely matched the cognitive demands of the multiple-choice assessment than the passive reading condition, potentially amplifying performance differences alongside retrieval and feedback effects. These features may help explain why retrieval practice with feedback supported learners' delayed performance on multiple-choice application tests. The present design was not intended to isolate the independent contribution of each component, but rather to evaluate a classroom-feasible version of retrieval practice with feedback for spelling-related knowledge. Whether these benefits extend to productive spelling or authentic writing remains a question for future research.

Addressing the unique challenges of EFL learners with non-alphabetic L1s

Our findings take on additional significance when considered within the specific context of Chinese learners acquiring English orthographic knowledge. The fundamental differences between Chinese logographic and English alphabetic writing systems create challenges for English spelling learning. Learners whose L1 involves character-based writing systems may approach English spelling through visual memorization strategies transferred from their first writing system—an approach that may be less well suited to English's morphophonological system (Wang et al., 2003). By requiring learners to actively generate and apply spelling patterns, retrieval practice with

feedback may encourage the analytical processing needed to work with systematic orthographic regularities.

The performance of our elementary school participants is also noteworthy given developmental considerations. These fourth-grade students were still consolidating their L1 literacy skills while developing L2 proficiency, a period in which cross-linguistic constraints may shape L2 reading and spelling development (Koda, 2007). In the present study, they were able to apply taught spelling rules and word-origin associations to novel items in multiple-choice application tests. This finding suggests that, under scaffolded classroom conditions, young EFL learners can engage with explicit spelling-related explanations. Whether such performance reflects broader metalinguistic development or developmental readiness for other forms of explicit instruction should be examined in future research.

Implications for delayed retention over one week

One of the notable findings across both experiments was the small observed change in the retrieval-practice-with-feedback condition over one week. Whereas performance in the reading condition showed a clearer decline, especially in Experiment 1, performance following retrieval practice with feedback showed little evidence of decline between the 24-h and one-week tests. This pattern may be relevant for EFL contexts where classroom time is limited and curricula are densely packed. Whether such performance patterns reduce the need for review or re-teaching should be examined in future classroom research. These interpretations, however, must be considered in light of the repeated testing design used in the present experiments, as performance at the one-week delay may have been influenced by prior exposure to the same items at the 24-h test.

The one-week retention pattern is consistent with the possibility that retrieval practice with feedback strengthened access to the target knowledge relative to reading-based study. Effortfully retrieving and applying the target knowledge, followed by corrective feedback, may have strengthened the cues available for later access to that knowledge (Carpenter, 2009). In addition, the desirable difficulty created by retrieval may have encouraged more effortful processing than passive reading (Bjork & Bjork, 2011). For spelling knowledge that students must maintain and use throughout their academic careers, improved memorability over time would represent a crucial advantage. However, the present behavioral data do not directly identify the underlying memory mechanisms, and the observed pattern may also reflect active application and the closer alignment between practice and assessment format. Thus, these mechanisms should be treated as plausible explanations for the observed one-week application-test performance rather than as mechanisms demonstrated directly by the present study.

Pedagogical implications for EFL spelling instruction

Our findings offer guidance for implementing the systematic spelling instruction of the kind advocated by some researchers (Joshi et al., 2008), but still relatively uncommon in L2 classroom practice. The retrieval practice with feedback approach

we tested potentially could be integrated into existing curricula without requiring extensive resources or teacher training. The key elements—initial instruction followed by application-based practice with feedback—align with communicative language teaching principles while maintaining the systematic coverage necessary for spelling-related knowledge.

For teachers accustomed to traditional approaches emphasizing copying and memorization, the present results suggest that application-based practice with feedback may be a useful complement to reading-based study. The initial discomfort that students may experience when generating answers, rather than simply rereading explanations, is consistent with the desirable difficulties framework, where conditions that make learning challenging can also be beneficial in the long term (Bjork & Bjork, 2011). The feedback component of our intervention also deserves attention. By providing explanations along with correct answers, we aimed to ensure that errors became learning opportunities rather than sources of frustration (Pan et al., 2020). This approach may be especially relevant in educational contexts where error-making can be viewed negatively. Whether such feedback also improves learners, error-related attitudes, motivation, or broader classroom achievement should be examined in future research (Metcalf, 2017; Roediger & Butler, 2011).

Etymology instruction: an underutilized resource

Experiment 2's focus on instructionally presented word-origin and word-structure associations addresses a particularly neglected aspect of L2 spelling instruction. Understanding word origins provides learners with a conceptual framework for making sense of English's seemingly irregular patterns (Devonshire & Fluck, 2010). The higher delayed multiple-choice application-test performance observed in Experiment 2 suggests that selected word-origin and word-structure associations can be incorporated into instruction for young EFL learners, at least when presented in the format used in the present study.

Word-origin knowledge may also be relevant to aspects of language learning beyond spelling, including vocabulary learning, reading comprehension, and metalinguistic awareness, although these outcomes were not directly assessed in the present study (Bristow, 2024; Newton, 2018). For EFL learners who will encounter increasing numbers of academic words with Greek and Latin origins as they advance in their English studies, future research should examine whether early exposure to such patterns provide broader preparation. Our results suggest that retrieval practice with feedback may be a classroom-feasible way to support delayed application-test performance for this targeted word-origin knowledge. It is important to note, however, that the utility of etymological instruction is inherently constrained by the extent to which target vocabulary contains transparent or teachable morphological and etymological structure. As such, its instructional benefits are likely to be most pronounced for word sets where such structure is present and identifiable and should not be interpreted as extending uniformly across all areas of vocabulary or spelling.

Limitations and future directions

Although our findings are encouraging, several limitations warrant consideration. As previously noted, the same content was tested on both retention tests in each experiment, which raises the prospect that retrieval practice effects from the first retention test may have influenced the second. We also relied on four-option multiple-choice application assessments, which are susceptible to guessing and do not assess productive spelling or authentic writing. The findings should therefore be interpreted as evidence for improved delayed application-test performance rather than direct evidence of gains in production-based spelling or writing. Future research should examine whether these benefits extend to tasks in which learners must generate complete spellings or use the target knowledge in authentic writing contexts.

Other limitations concern differences in instructional and assessment design across conditions. Retrieval practice in this study required active generation and application of spelling rules with corrective feedback, whereas reading involved passive study only. Moreover, as previously noted, the retrieval tasks were also more closely aligned with the retention test format (i.e., rule application to novel items), consistent with transfer-appropriate processing principles (Morris et al., 1977). Because the retrieval-practice condition differed from reading in several respects—retrieval, corrective feedback, active application, and closer alignment with the assessment format—the findings should not be interpreted as isolating a pure retrieval-practice effect. Instead, they evaluate a classroom-feasible, application-based retrieval-practice-with-feedback approach. Future research should separate these components more directly to estimate the relative contributions of retrieval, feedback, application demands, and practice–assessment alignment.

We also used simplified etymological concepts that occasionally deviated from strict historical-linguistic etymology, and to maintain developmental appropriateness, some explanations combined classical etymological relations with general semantic or associative information. Hence, our findings reflect the acquisition of instructionally presented etymological knowledge rather than exhaustive historical-linguistic reconstructions. Future research could use more extensively validated instructional materials that explicitly distinguish among classical combining forms, affixes, lexical bases, place-name elements, and related semantic associations.

Developmental factors—such as cognitive load and working memory demands associated with processing explicit rule-based explanations—may also have influenced the present results. Future work could address those and other factors such as prior English proficiency, working memory capacity, or L1 literacy skills (cf., Pan et al., 2025; Smith-Peirce & Butler, 2025; Terai et al., 2021). To clarify the robustness and optimal implementations of retrieval practice for spelling acquisition, researchers could investigate longer retention intervals (e.g., Carpenter, 2009), distribute practice across multiple sessions to leverage spacing effects (e.g., Fang et al., 2024; Nakata et al., 2021; Rogers, 2015; Sonbul & Siyanova-Chanturia, 2024; Suzuki & DeKeyser, 2017), and explore alternative practice formats (e.g., Puimège et al., 2025).

Broader implications for L2 instruction

Although we focused on spelling, our findings may inform future research on retrieval-practice-based instruction for other forms of rule-based L2 knowledge. Many aspects of language learning involve rule-based systems that learners must apply, including grammatical structures, pronunciation patterns, pragmatic conventions, and morphological processes. The present findings do not establish transfer to these domains, but they suggest that application-based retrieval practice with feedback may be worth testing in them.

Our results also contribute to ongoing discussions about explicit versus implicit instruction in L2 contexts, though within the specific domain of spelling-related knowledge and multiple-choice application-test performance. Although communicative approaches often de-emphasize explicit rule teaching, our findings suggest that explicit instruction followed by active application and corrective feedback can support learners' delayed application of recently taught spelling-related knowledge. Whether this approach supports procedural knowledge or more fluent use in production-based tasks remains a question for future research.

Conclusions

The present study shows that retrieval practice with feedback can improve elementary EFL learners' delayed performance on multiple-choice application tests of spelling-rule and word-origin knowledge. By examining morphophonological rules and instructionally presented word-origin associations, this study extends retrieval-practice research to two forms of spelling-related knowledge that require learners to apply recently taught information to novel items. The findings suggest that retrieval practice with feedback may be useful for supporting delayed application of recently taught spelling-related knowledge, as assessed by multiple-choice application tests, although further research is needed to determine whether these benefits extend to other measures of learning and longer retention intervals.

For practitioners, our findings point to a classroom-feasible complement to traditional rote memorization approaches in EFL spelling instruction. The retrieval-practice-with-feedback procedure tested here requires no special materials or technology, which may make it practical for ordinary classroom use. More broadly, the findings illustrate how principles from cognitive science may inform the design of L2 spelling instruction, while also pointing to the need for future work using production-based spelling, writing, and longer-term classroom outcomes.

As English continues to serve as a global lingua franca, developing effective methods for teaching its orthographic system becomes increasingly important. For Chinese EFL learners and other learners facing similar challenges, retrieval practice with feedback may offer a classroom-feasible way to support delayed application-test performance for spelling-related knowledge. Future work should examine whether these benefits extend to productive spelling, authentic writing, and longer curriculum-based learning.

Appendix A

Spelling rules used in Experiment 1

No	Name of rule	Definition	Examples
1	Add <i>-ing</i> directly	For most verbs, simply add <i>-ing</i> to form the progressive tense without changing the base spelling	<i>go</i> → <i>going</i> ; <i>drink</i> → <i>drinking</i>
2	Double final consonant before <i>-ing</i>	If a verb ends in a short vowel followed by a single consonant, double the consonant before adding <i>-ing</i>	<i>run</i> → <i>running</i> ; <i>stop</i> → <i>stopping</i>
3	Drop silent <i>e</i> before <i>-ing</i>	If a verb ends in silent <i>e</i> , drop the <i>e</i> before adding <i>-ing</i>	<i>come</i> → <i>coming</i> ; <i>write</i> → <i>writing</i>
4	Add <i>-s</i> or <i>-es</i> for plurals	Add <i>-s</i> to form plurals for most nouns. If the word ends in <i>-o</i> , add <i>-es</i> if animate, <i>-s</i> if inanimate	<i>pear</i> → <i>pears</i> ; <i>hero</i> → <i>heroes</i> ; <i>zoo</i> → <i>zoos</i>
5	<i>i</i> before <i>e</i> except after <i>c</i>	When <i>i</i> and <i>e</i> appear together, spell <i>ie</i> unless it follows <i>c</i> , in which case spell <i>ei</i>	<i>believe</i> ; <i>receive</i>
6	Prefix <i>en-</i> means “to make...”	Words beginning with <i>en-</i> often mean “to make [adjective/noun].”	<i>enrich</i> ; <i>enjoy</i> ; <i>enlarge</i>
7	<i>-dge</i> spelling after short vowel	After a short vowel sound, the /dʒ/ sound is typically spelled <i>dge</i> at the end of a word	<i>judge</i> ; <i>bridge</i>
8	<i>c</i> spelling rule	The letter <i>c</i> spells /k/ before <i>a</i> , <i>o</i> , <i>u</i> , and /s/ before <i>e</i> , <i>i</i> , <i>y</i>	<i>cat</i> , <i>cake</i> ; <i>cent</i> , <i>city</i> , <i>cycle</i>
9	Add <i>-ed</i> directly	For regular past-tense verbs, add <i>-ed</i> to the base form without altering the original spelling	<i>help</i> → <i>helped</i> ; <i>show</i> → <i>showed</i>
10	Change <i>y</i> to <i>i</i> and add <i>-ed</i>	If a verb ends in a consonant + <i>y</i> , change the <i>y</i> to <i>i</i> before adding <i>-ed</i>	<i>study</i> → <i>studied</i> ; <i>cry</i> → <i>cried</i>
11	Add <i>-d</i> to words ending in <i>e</i>	If a verb ends in <i>e</i> , add <i>-d</i> to form the past tense (do not double the <i>e</i>)	<i>use</i> → <i>used</i> ; <i>live</i> → <i>lived</i>
12	Change <i>f/fe</i> to <i>ves</i> in plurals	For some nouns ending in <i>f</i> or <i>fe</i> , change the ending to <i>ves</i> in the plural form	<i>knife</i> → <i>knives</i> ; <i>leaf</i> → <i>leaves</i>
13	<i>q</i> is followed by <i>u</i>	The letter <i>q</i> is almost always followed by <i>u</i>	<i>queen</i> ; <i>quiet</i> ; <i>quick</i>
14	Add <i>-est</i> to form superlatives	For many short adjectives, add <i>-est</i> to the base word to form the superlative	<i>small</i> → <i>smallest</i> ; <i>old</i> → <i>oldest</i>
15	<i>x</i> spelling pattern	The /ks/ sound is commonly spelled with <i>x</i> in English words	<i>box</i> ; <i>six</i> ; <i>fox</i>
16	Silent <i>e</i> for long <i>a</i> (a-e pattern)	The long <i>a</i> sound is often spelled <i>a-consonant-e</i> , with the <i>e</i> being silent	<i>cake</i> ; <i>name</i> ; <i>face</i>
17	Add <i>-ing</i> to verbs ending in <i>ie</i>	Change <i>ie</i> to <i>y</i> before adding <i>-ing</i>	<i>die</i> → <i>dying</i> ; <i>lie</i> → <i>lying</i>
18	Add <i>-es</i> to words ending in <i>-ch/sh/x/s/z</i>	For nouns ending in these sounds, add <i>-es</i> to form the plural	<i>box</i> → <i>boxes</i> ; <i>bush</i> → <i>bushes</i> ; <i>class</i> → <i>classes</i>
19	Silent <i>k</i> before <i>n</i>	In words beginning with <i>kn</i> , the <i>k</i> is silent	<i>know</i> ; <i>knife</i>
20	Silent <i>w</i> before <i>r</i>	In words beginning with <i>wr</i> , the <i>w</i> is silent	<i>write</i> ; <i>wrong</i>

Appendix B

Etymological concepts used in Experiment 2

No	Element/Root	Definition	Examples
1	<i>bio-</i>	Greek root meaning “life” or “living.”	<i>biology,</i> <i>biography</i>
2	<i>nat-</i>	Latin root related to “birth” or “being born.”	<i>native, nature</i>
3	<i>hum-</i>	Latin-derived element related to humus (“earth” or “soil”), as seen in words associated with lowness or the earth	<i>humility,</i> <i>humble</i>
4	<i>mus-</i>	Greek-derived element from Mousa / the Muses, goddesses associated with the arts and inspiration	<i>music, museum</i>
5	<i>geo-</i>	Greek root meaning “earth,” from goddess Gaia	<i>geology,</i> <i>geography</i>
6	<i>phon-</i>	Greek combining form meaning “sound” or “voice.”	<i>telephone,</i> <i>microphone,</i> <i>symphony</i>
7	<i>tele-</i>	Greek combining form meaning “far” or “distant.”	<i>telephone,</i> <i>television</i>
8	<i>dog-</i>	English base word used in compounds and derived words, often carrying cultural associations such as loyalty or persistence	<i>dogged, dog</i> <i>tag</i>
9	<i>cat-</i>	English base word used in derived words and idiomatic expressions, sometimes carrying cultural associations such as slyness or sharpness	<i>catty, cat’s paw</i>
10	<i>-land</i>	English/Germanic place-name element meaning “land” or “territory associated with a person or group.”	<i>England,</i> <i>Queensland</i>
11	<i>-ton</i>	English place-name element from Old English tūn, meaning “enclosure,” “settlement,” or “town.”	<i>Boston,</i> <i>Princeton</i>
12	<i>-tion</i>	Latin-derived noun-forming ending, often entering English through French, indicating an action, process, or state	<i>action, atten-</i> <i>tion, emotion</i>
13	<i>re-</i>	Latin-derived prefix, often entering English through French or Anglo-French, meaning “again,” “back,” or “backward.”	<i>rebuild, restart</i>
14	<i>-ee</i>	French/Anglo-French-derived suffix often indicating the person who receives or undergoes an action	<i>employee,</i> <i>appointee</i>
15	<i>sci-</i>	Latin root meaning “to know” or “knowledge.”	<i>science, scien-</i> <i>tist, conscious</i>
16	<i>cyber-</i>	Greek-derived combining form originally related to “steersman” or “governance,” now associated with computers, networks, and digital environments	<i>cyberpunk,</i> <i>cyber</i>

The elements presented in this appendix are instructional etymological groupings used to support learning of word structure in English, rather than exhaustive or fully reconstructed historical derivations. Entries combine classical roots, affixes, and selected lexical stems, grouped by shared origin or conventional historical association in a simplified pedagogical form. Some items reflect historically derived forms (e.g., Latin or Greek roots and affixes), while others represent lexical bases included for instructional consistency within the teaching materials

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